



FIRST SEMESTER 2026-27 COURSE HANDOUT

Date: 01/08/2026

In addition to Part I (General Handout for all courses appended to the Time Table), this part gives further specific details regarding the course.

Course No. : MATH F311

Course Title : INTRODUCTION TO TOPOLOGY

Instructor-in-charge : SHUCHITA GOYAL (shuchita.goyal@pilani.bits-pilani.ac.in, 6166-B)

1. Course Description: Topology and topological spaces, Metric spaces, Product and Quotient topology, Continuity and Homeomorphism, Connectedness, Compactness, Normal spaces, Classification of compact surfaces.

2. Scope and Objective of the Course: To introduce the students to the concepts of logical thinking in abstract terms using formal and axiomatic methods and to lay the foundations for further studies in abstract Mathematics.

3. Course learning outcomes:

Upon successful completion of this course, students should be able to:

CLO1. Define and analyze basic topological concepts, including topologies, bases, subspace topologies, product topologies, and quotient topologies, to construct and classify fundamental topological spaces.

CLO2. Evaluate continuity and homeomorphisms to determine topologic equivalence between abstract spaces and investigate topological invariants.

CLO3. Demonstrate a rigorous understanding of topological properties such as connectedness, compactness, countability, and separation axioms (including Hausdorff and Normal spaces).

CLO4. Apply and construct formal mathematical proofs utilizing key structural theorems, including Urysohn's Lemma, Urysohn Metrization Theorem, and Tietze Extension Theorem.

4. Textbooks and references:

TB1. Munkres, J. R.: *Topology*, Pearson Education, 2000 (2nd Edition)

RB1. Armstrong, M.A.: *Basic Topology*, Springer UTM, 1983.

RB2. Adams C. and Franzosa R.: *Introduction to Topology Pure and Applied*, Pearson Education, 2009 (1st Edition)

RB3. Simmons G. F.: *Introduction to Topology and Modern Analysis*, Tata McGraw Hill, 2004.

5. Course Format and Weekly Workflow:



Birla Institute of Technology & Science, Pilani

Pilani | Dubai | Goa | Hyderabad

ACADEMIC— UNDER GRADUATE STUDIES DIVISION

Module	Lecture Session	Reference	Learning Outcome
1. Revision of Metric Spaces, Definition of topology and elementary examples of topological spaces.	1-4	Chapter 2 Sec. 12-13	Topological Spaces, Basis for a topology.
2. Constructing new topological spaces from a given topological space.	5-9	Chapter 2 Sec. 14-16	Order topology, Product topology on $X \times Y$, Subspace topology.
3. Hausdorff axiom.	10-11	Chapter 2 Sec. 17	Closed sets, Limit point, Hausdorff spaces.
4. Continuity and properties of continuous functions.	12-15	Chapter 2 Sec. 18	Continuous functions and Homeomorphisms.
5. Methods for imposing topologies on sets.	16-20	Chapter 2 Sec. 19-22	The Product topology, The Metric topology and The Quotient topology.
6. Connectedness and its importance.	21-25	Chapter 3 Sec. 23-25	Connected spaces, Connected Subspaces of Real Line, Components
7. Compactness and its importance.	26-30	Chapter 3 Sec. 26-29	Compact spaces, Compact subspaces of Real Line, Limit point compactness, Local Compactness
8. Countability & Separation Axioms.	31-40	Chapter-4 Sec. 30-35	First countable spaces, Second countable spaces, Regular spaces, Normal Spaces, The Urysohn Lemma, The Urysohn Metrization Theorem, The Tietze extension theorem.

6. Evaluation Scheme:

Components	Duration	Weightage	Date & Time	Nature of component (Close Book/ Open Book)
Quizzes*	20 Min.	30% (60 Marks)	Announced	Closed Book
Mid-Semester Exam	90 Min.	30% (60 Marks)	As announced by AUGSD	Closed Book
Comprehensive Examination	180 Min.	40% (80 Marks)	As announced by AUGSD	Closed Book+Open Book

*There will be four announced class tests of 20 marks and 20 minutes each in lectures/tutorials. Best



Birla Institute of Technology & Science, Pilani

Pilani | Dubai | Goa | Hyderabad

ACADEMIC— UNDER GRADUATE STUDIES DIVISION

three among them will be considered for evaluation. All the class tests will be offline and no electronic device such as laptop, tablet, mobile phone etc. will be allowed in any circumstances.

- 7. Make-Up Policy:** Makeup applications for the mid semester and comprehensive exams will be approved only on presenting suitable documents. **No makeup for class tests will be given under any circumstances.**
- 8. Attendance Policy:** Each student is expected to attend all classes and appear in all evaluative components.
- 9. Cut off for NC grade:** Any student scoring less than 10% of the total marks (i.e., less than 20 marks out of 200) will be given an NC grade. Further, if a student is absent in mid-semester or comprehensive exam or in all four quizzes, then the student will be given an NC grade.
- 10. Chamber Consultation Hour:** Every Monday 4-5 pm.
- 11. Notices:** All notices concerning this course will be put up only on NALANDA.

Instructor – in charge
MATH F311